

Statistical Estimation for BGG Ratings

Liam Whitenack
DASC 605
Professor Trent Buskirk
Old Dominion University

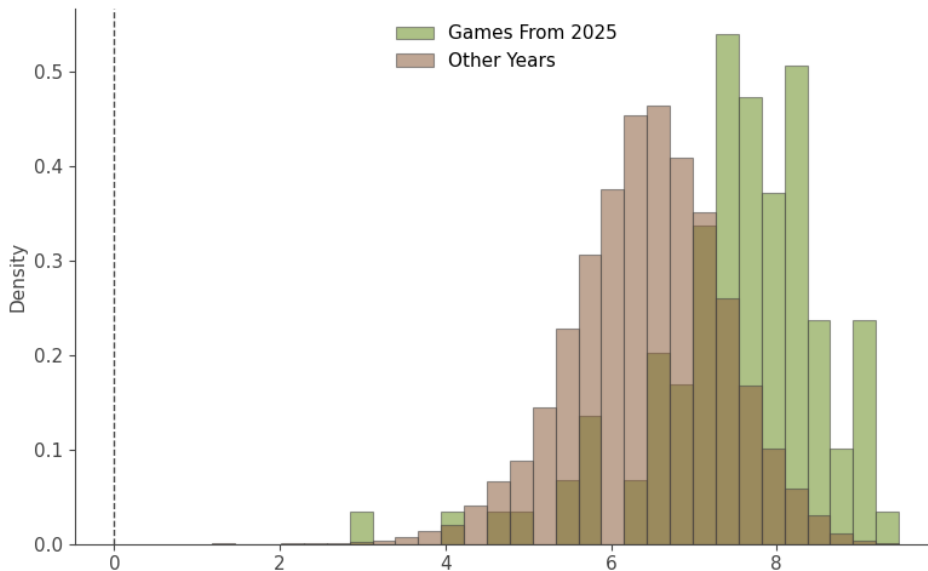
March 5, 2026

- Collected in August 2025 and March 2026
- Source: [BoardGameGeek XML API](#)
- Processing code: [GitHub Repository](#)
- A much more minimal copy can be downloaded by pressing this [link](#)

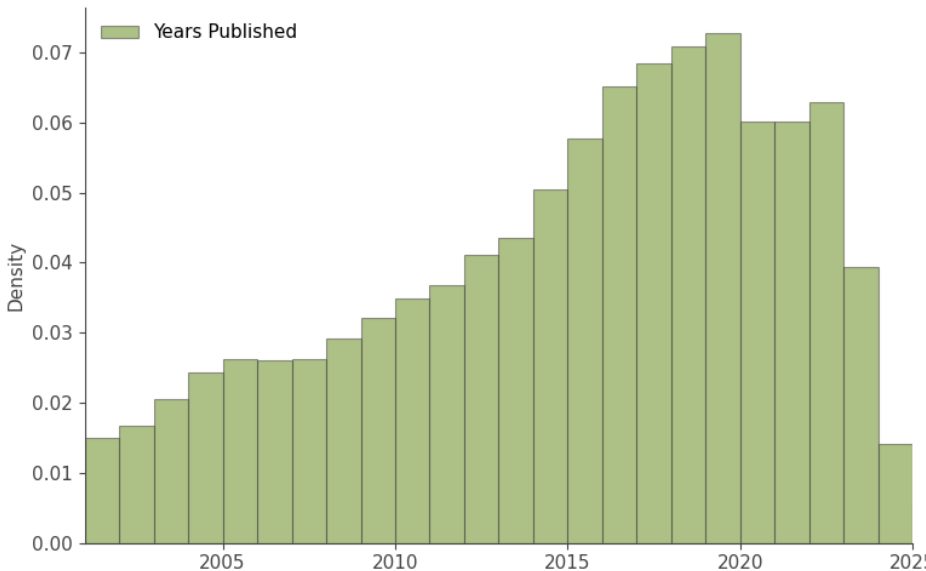
Fetching Game Data from BoardGameGeek

```
def fetch_things(
    ids: Iterable[int], max_retries: int, retries: int
) -> list[dict]:
    url = f"{'url'}/thing?id={','.join(map(str, ids))}&stats=1"
    retries = 0
    while retries < max_retries:
        response = requests.get(url)
        if response.status_code == 200:
            return parse(response.text)["items"]["item"]
        else:
            time.sleep(5)
    raise Exception()
```

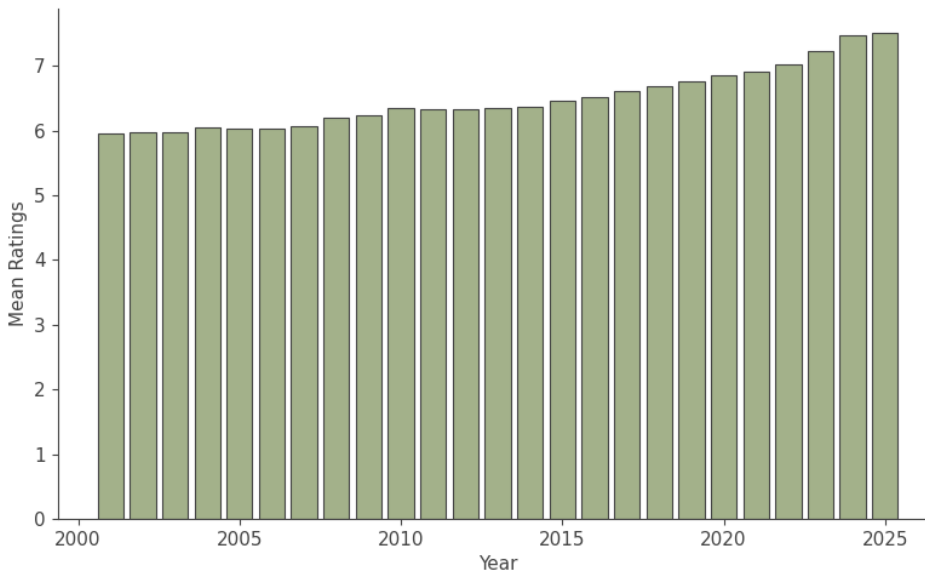
Distribution of Ratings



Games Published Per Year



Mean Rating by Year



Confidence Interval for Mean Rating

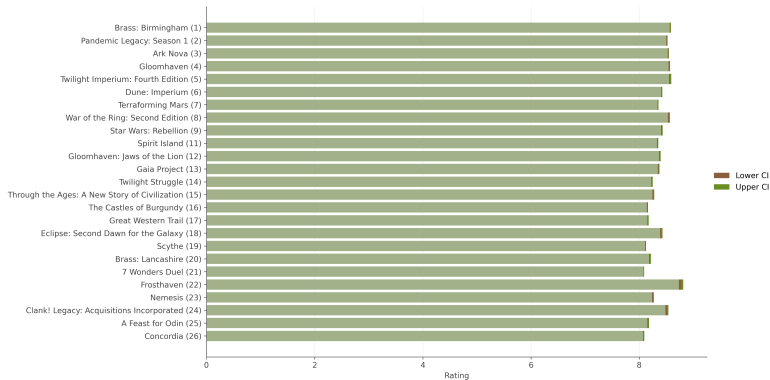
Goal: Estimate a confidence interval for the true mean rating of a game.

- Uses the **t-distribution** because the population standard deviation is unknown.
- Standard error is computed from the sample ratings.
- Degrees of freedom: $df = n - 1$

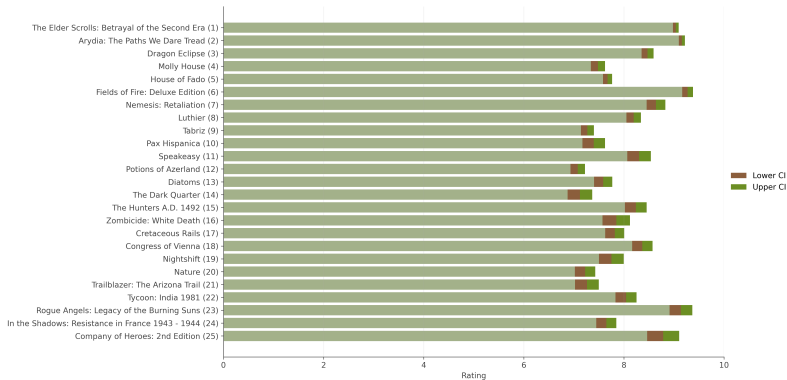
Implementation

```
se = self._standard_error(df = self.usersrated - 1) t_crit = stats.t.ppf((1 + confidence)/2, df)
margin = t_crit * se
return(self.average - margin, self.average + margin)
```

Top 25 Games



Top 25 Games vs. 2025 Releases



Updated 2025 Comparison

